```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

use IEEE.NUMERIC_STD.ALL;



entity mod6_counter is

  Port (

      clk : in STD_LOGIC;

      rst : in STD_LOGIC;

      q : out STD_LOGIC_VECTOR(3 downto 0)
  );

end mod6_counter;



architecture Behavioral of mod6_counter is



  signal count : unsigned(3 downto 0) := "0000";
```

```
 signal clk_count : integer range 0 to 49999999 := 0;


begin



 process(clk, rst)

 begin



    if rst = '1' then
      count <= "0000";
      clk_count <= 0;



    elsif rising_edge(clk) then
    if clk_count = 49999999 then

      clk_count <= 0;


      if count = 5 then

          count <= "0000";
```

```
        else

            count <= count + 1;

        end if;



      else

        clk_count <= clk_count + 1;
      end if;



   end if;



 end process;



 q <= std_logic_vector(count);



end Behavioral;
```